

Cross-Layer Design in the Internet of Things (IoT): Issues and Possible Solutions

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Abstract-Connectivity has been transformed by the Internet of Things (IoT), which has made possible an extensive network of interconnected devices. Despite this, cross-layer design poses considerable obstacles due to the complexity of IoT systems. In this study, several solutions to the problems with cross-layer design on the Internet of Things are discussed. The analysis highlights how critical cross-layer design is to maximize system efficiency, reduce energy use, and guarantee effective resource management. Common issues are covered, such as data security, network efficiency, and interoperability issues. The study also highlights how important cross-layer design is for ensuring secure and dependable connectivity in Internet of Things devices. It suggests defenses against possible attacks on IoT devices and data. This study advances the Internet of Things (IoT) ecosystem by providing insights into cross-layer design difficulties and possible solutions, opening the door for more effective, secure, and networked smart technology.

Index Terms – Cross-Layer Design, Internet of Things (IoT), smart technology

I.INTRODUCTION

The Internet of Things (IoT) promises a linked world where systems, sensors, and gadgets seamlessly interact to improve productivity, effectiveness, and the standard of life. The expansion of IoT devices over the

last ten years has had a profoundly disruptive influence on a wide range of businesses, economies, and communities, including manufacturing, transportation, healthcare, and agriculture [1]. Technological developments in data analytics, cloud computing, wireless communication, and hardware shrinking have all contributed to this paradigm shift toward networked smart devices [1]. Figure 1 shows the significant benefit of IoT technologies lies in their connectivity feature, which holds immense promise [1]. Due to this, the Internet of Things (IoT) landscape has expanded into a massive ecosystem that includes billions of networked devices, ranging from smart infrastructure and environmental sensors to consumer electronics and industrial gear [2].

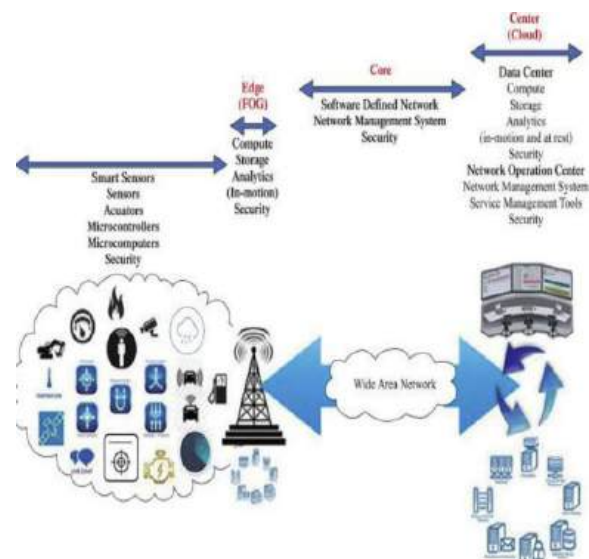


Figure 1. Standard network architecture and interconnectivity of the IoT [1]

But despite all the potential of IoT, there remains a complicated web of opportunities and problems, especially when it comes to cross-layer design. The deliberate integration and optimization of capabilities, interactions, and communication protocols across several levels of the Internet of Things architecture is known as cross-layer design [3]. In contrast to conventional layered protocol designs, which divide functionalities into discrete layers, cross-layer design aims to maximize interlayer dependencies and synergies to improve system scalability, performance, and energy efficiency [4]. In TCP/IP networks, the collaboration of all five layers achieves the end-to-end connection [5]. To create and preserve device communication, these layers cooperate. There are not many interactions between different levels in their design. This cooperation is a key component of the TCP/IP architecture, which creates a system that resembles a black box by restricting data and service calls to nearby levels [5].

The intrinsic variety, scalability, and dynamic nature of IoT networks make cross-layer design essential [6]. Each layer in conventional layered systems, like the OSI model, carries out certain tasks and interacts with neighboring levels via defined interfaces. However, the strict layer boundaries of conventional designs can stifle creativity and efficiency in Internet of Things settings, which are defined by a diversity of devices, communication protocols, and application domains [6]. Cross-layer design is an approach to networking that aims to enhance network efficiency and reliability through the interaction of multiple OSI or TCP/IP model layers [7].

By facilitating coordination and communication across non-adjacent levels, cross-layer architecture provides a flexible and adaptable method to handle these issues [8]. This results in better resource allocation,

increased security, and improved Quality of Service (QoS) [9]. Strong cross-layer design is becoming more and more necessary as IoT applications spread over a variety of sectors, from industrial automation and environmental monitoring to smart cities and households [9]. IoT systems must make their way through a variety of environmental conditions, dynamic network topologies, and demanding performance standards [9]. Examples of these environments range from sensor networks placed in agricultural areas to smart grids controlling energy distribution [10]. One of the many difficulties when applying cross-layer design in Internet of Things networks involves managing competing design goals, such as network dependability and energy efficiency. Significant challenges include ensuring safe data transfer, enabling interoperability among various IoT devices, and maximizing data routing [11].

An extensive amount of research is being done to investigate innovative cross-layer design topologies, protocols, and algorithms that are specific to the needs of IoT ecosystems considering these constraints [12]. Through the integration of knowledge from several fields including machine learning, embedded systems, and networking, researchers want to create cross-layer design solutions that are safe, scalable, and able to power the next wave of the Internet of Things. We present a thorough investigation of cross-layer design in the Internet of Things in this article. We explore the foundations of cross-layer design, looking at its ideas, practices, and uses in various Internet of Things scenarios.

II. INTERNET OF THINGS

A. Architectural Considerations in IoT

The Internet of Things (IoT) functions according to a complex architectural framework that has been carefully crafted to enable smooth integration, exchange of data, and compatibility among a wide range of devices and systems [13]. We will further examine the architectural paradigms suggested for the effective operation and development of IoT ecosystems [13].

B. Architecture overview

IoT system architecture is fundamentally the blueprint that controls how virtual networks and physical items interact with one another [14]. The Internet of Things (IoT) was first conceptualized in 1999 by Kevin Ashton, who saw an opportunity for networked devices to use data to maximize resource consumption, improve decision-making, and spur innovation across sectors [14].

C. Layered Architectures in IoT

a. Three-Layer Architecture

Numerous Internet of Things implementations are built upon the cornerstone of the traditional three-layer architecture [15]. The perception layer serves as the main interface and uses a variety of sensors that are built into actual items. The Network Layer serves as an intermediary, enabling safe data transfer over a variety of communication methods, such as wired and wireless networks [15]. In addition to other connectivity choices, the network tier must support secure wired and wireless connections. The application layer accepts data, processes it, and analyzes it to provide operations and make choices. It will then use the network layer to send the results up to the perception layer [15]. Figure 2 shows the three-layer architecture of IoT [16].

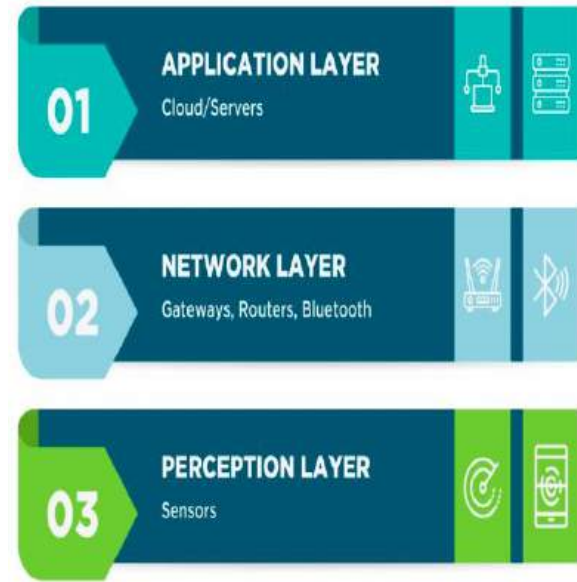


Figure 2. 3-layered IoT Structure [16]

b. Four Layer Architecture

Expanding upon the fundamental three-layer framework, scholars and professionals have suggested an expanded four-layer structure in order to tackle emergent complexities and challenges in the Internet of Things deployments [17]. The Support Layer, which is an additional stratum, is specifically designed to enhance mechanisms for data security, privacy, and authentication. Within the IoT ecosystem, the Support Layer protects against fraudulent attempts and unauthorized access by utilizing strong authentication frameworks, access control mechanisms, and encryption protocols in place to guarantee the integrity and confidentiality of data flows [17] [4]. Figure 3 shows the four-layer architecture of IoT [16].



Figure 3. 4-layered IoT Structure [16]

c. Five-Layer Architecture

The five-layer architecture, which includes the Middleware Layer and Presentation Layer to further improve functionality and user experience, signifies the zenith of IoT architectural sophistication [18]. The Middleware Layer, alternatively referred to as the Processing Layer, functions as a computational engine, utilizing sophisticated algorithms and data processing methods to derive significant insights from the extensive collections of IoT data [18][4]. To bridge the gap between complicated IoT systems and end-user demands, the Presentation Layer concentrates on developing user-friendly apps and user interfaces. The Presentation Layer enables consumers to fully utilize IoT technology and spur innovation across a range of industries with engaging user experiences, interactive dashboards, and real-time analytics [18]. Figure 4 shows the five-layer architecture of IoT [16].

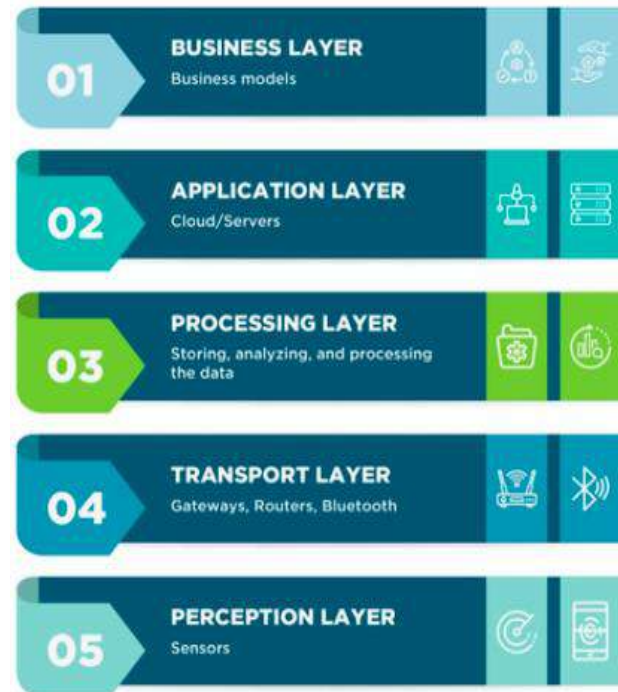


Figure 4. 5-layered IoT Structure [16]

III. CROSS-LAYER DESIGN

The implementation of cross-layer design in wireless communication systems holds great potential in terms of enhancing data transfer rates and optimizing overall network performance [19]. Significant research endeavors have been focused in recent years on the development of alternative architectures, which deviate from the conventional layered model. The primary objective of these endeavors is to decrease interdependencies between layers and improve the overall efficiency of the system [19].

An inherent difficulty in wireless networks pertains to the efficient administration of interactions among the protocol stack, including but not limited to the physical, data-link, network, transport, and application layers [20]. Cross-layer architecture aims to mitigate these obstacles through the facilitation of information and

communication exchange among numerous layers, consequently enhancing vital components including mobility, security, and Quality of Service (QoS) [20].

One possible way to increase the smart grid's efficiency is through the incorporation of sensor networks, control of energy, and connection networks via the Internet of Things [21]. Continuous monitoring of energy output and consumption is made feasible by the integration of sensors into the grid structures, which improves resource management. Furthermore, communication technologies facilitate the smooth interchange of information across different grid components, including substations, power plants, and residential buildings, improving coordination and optimizing the management of energy assets [21].

A. Security:

In current network environments, the implementation of strong security protocols is critical in light of the escalating complexity of cyber threats [22]. Although conventional layered architectures are successful in structuring network protocols, they frequently encounter difficulties in sufficiently mitigating security issues [22].

Cross-layer architecture, which permits information sharing and communication between several protocol stack layers, offers a viable method for improving network security [23]. Encryption solutions like end-to-end encryption and Wi-Fi-protected access enable the establishment of secure communication channels by enabling cross-layer interactions [23]. This method lowers the possibility of unwanted access or interruptions to data security while simultaneously improving data confidentiality and integrity [23].

Enabling collaboration between security protocols at various layers, cross-layer architecture additionally facilitates more

effective intrusion prevention and detection mechanisms [24]. Intrusion detection systems (IDS) that are implemented at the network layer have the ability to utilize data from the data-link layer in order to detect and address possible security risks in advance [24].

B. Quality of Service (QoS):

In order to overcome difficulties, cross-layer architecture provides more adaptable interactions across layers, facilitating adaptive QoS control and dynamic resource allocation. Figure 5 shows how IoT provides QoS [23]. Cross-layer optimization improves the network's capacity to prioritize important data streams, adjust to shifting traffic circumstances, and lessen congestion by enabling information interchange across layers [25] [4].

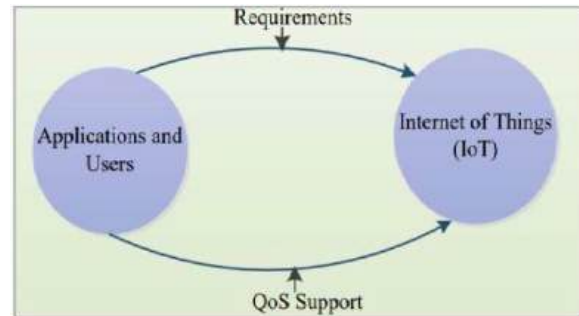


Figure 5. demonstrates how to provide Internet of Things gadgets with Quality of Service (QoS) [23]

Furthermore, complex QoS mechanisms like packet scheduling, admission control, and traffic shaping may be implemented at different protocol stack tiers due to cross-layer architecture [26]. For example, the system may prioritize applications that run in real-time, handle bandwidth allocation more effectively, and provide a consistent user experience by synchronizing QoS policies among the network and transport layers [26].

In addition, QoS parameters may be more precisely controlled due to cross-layer

optimization, which gives network managers the ability to customize performance measurements to meet the needs of certain applications [27]. In particular, in critical applications like a telemedicine driverless car, and manufacturing automation, this method increases network resilience, lowers latency, and enriches the user experience as a whole [27].

C. Mobility:

In wireless communication networks, where equipment may need to roam across various network zones while maintaining uninterrupted connectivity, mobility management is an essential factor to take into account [9][4]. The insufficient coordination and sharing of information among levels in traditional layered designs frequently makes it difficult to provide smooth mobility [9].

In order to overcome these obstacles, cross-layer design promotes cooperation and communication across levels, opening the door to more effective mobility management techniques [28]. Cross-layer optimization facilitates smooth handovers, reduces service interruptions, and maximizes resource efficiency by exchanging data on device mobility and network conditions across layers [28].

IV. CROSS LAYER DESIGN IN IOT

Enabling collaboration and communication between protocol stack layers, cross-layer design in communication systems, such as IoT devices, enhances dependability and performance [23][4]. In contrast to conventional independent operation, cross-layer design endeavors to optimize system performance through the facilitation of interaction among tiers [29]. The issues presented by IoT devices, such as few

resources and heterogeneous application requirements, must be addressed with this strategy. Cross-layer design in IoT devices has several important advantages and characteristics [6][4].

It has been determined that cross-layer designs significantly improve wireless networks [29]. Cross-layer design is the process of facilitating communication between any two or more adjacent or non-adjacent layers in the protocol stack. Cross-layer design endeavors to optimize network performance by augmenting data transfer rates and elevating service quality by subverting the conventional division across layers in a wireless network's architecture [29]. Conventional wireless network layered design, which is characterized by stringent layer separation, can result in interdependency problems and inefficiencies within layers. By promoting communication and cooperation between various network stack tiers, cross-layer architecture tackles these issues. [29]

The Internet of Things (IoT) is a fast-growing field with many potential applications. Because cross-layer design may hide information, it is especially well suited for these kinds of applications. In the IoT space, cross-layer communication has several benefits [30]. As a preventive measure, one researcher presented a distributed cross-layer authentication design that makes use of physical layer authentication. A further study examined [31] security functions and threats at the device, network, and service levels and suggested a cross-layer framework to link and integrate these layers.

As a result, performance and resource utilization are typically suboptimal and fail to fully capture the complex interdependencies and interconnections across several levels. Identifying possible synergies and trade-offs

across layers and taking into account their interactions is how cross-layer design tackles this difficulty [32]. Optimization of physical layer characteristics, such as modulation methods and transmission power, in conjunction with network layer routing strategies, for instance, can result in notable energy savings [32].

V. CROSS-LAYER DESIGN PROBLEMS AND SOLUTIONS IN IOT DEVICES

The vast breadth and quick expansion of the Internet of Things (IoT) can be attributed to the inclusion of a wide range of devices. IoT applications improve system performance by utilizing cross-layer design, which provides efficiency and ease [6]. Nevertheless, managing the wide range of networked devices on the Internet of Things and putting cross-layer design into practice present difficulties. These difficulties include constant power consumption, device mobility, lack of standards, and security flaws. Although answers are put out, they are not set in stone and might change as technology develops [33]. Below are some of the issues and the solutions of IoT in cross Layer Design:

Security Issues:

The pervasive nature of security threats throughout the architecture of the Internet of Things (IoT) is further compounded by the vast interconnectivity of devices. Denial of Service (DoS) assaults present a substantial peril across all tiers, impeding authorized users' access to devices, resources, or networks [34][35]. In addition, limitations regarding storage capacity, processing speed, and power intensify vulnerability to a wide range of threats [34][35].

Concerning authentication mechanisms and data privacy, application security issues are

exacerbated by insufficient global regulations and standards. Service availability is jeopardized by the strain on data-analysis infrastructure caused by the overwhelming number of interconnected devices [34][35].

It is still quite difficult to put strong security measures in place when developing and deploying IoT devices. Devices with low memory and computing power are more susceptible to manipulation, illegal access, and data breaches because they cannot implement robust security measures [36]. Hackers can take advantage of flaws including obsolete firmware, poor encryption, and insufficient passwords. Due to their variety, it is difficult yet necessary to standardize security procedures across the vast array of IoT devices. Security issues therefore cover many tiers of IoT systems in addition to the external environment [37].

Unauthorized access and data manipulation are possible on devices due to inadequate verification, encryption, and communication protocols. These vulnerabilities are made worse by inadequate device and software management [38].

Security Solutions:

Hashing and feature extraction are used to verify users across platforms and terminal nodes in a mutual authentication solution for the Internet of Things that was suggested in 2011 [35]. To prevent collision attacks, feature extraction and hash function were merged to provide a dependable authentication option for IoT. The feature extraction method used is lightweight and irreversible, which is crucial for Internet of Things applications [35]. Still, the technique focuses on authentication during platform-to-terminal-node data transmission. The solution is theoretical and lacks empirical support, even though it has the potential to

improve security and decrease data transmission [35].

Mobility Issues:

Mobility offers several advantages, including greater adaptability, the addition of new services, and a broadening of IoT application areas [39]. Since node disconnections and irregular connections cause data loss and transmission delays, it is imperative to address these issues [40]. It is crucial to effectively manage mobility by rapidly connecting to mobile nodes to lessen these problems. As a result, even in the event of node movement, operation is made possible by constant connectivity and communication. While an abundance of studies has addressed handoffs in general, very few have particularly addressed the difficulties presented by high-speed train situations. High-speed rail moves quickly, which means that handoffs occur often. This can have a major impact on CBTC efficiency [41].

Mobility Solutions:

Using decision logic at the network and interface levels is necessary to properly handle mobility issues. Improved communication across layers is achieved by customizing solutions for each layer as opposed to using a general strategy. While some layers work well on their own to handle mobility, others need to work in tandem with other levels [42]. The use of cross-layer network concepts is necessary to put ideal solutions to mobility issues into practice [42]. Identifying link layer signals before and following handoff can further improve mobility by enabling communication via a cross-layer architecture [42]. The location of the mobile node and the positions of neighboring access points are often included in the messages that are exchanged during this procedure, which effectively directs traffic to reduce congestion [43].

Energy-efficiency Issues:

Although low-power devices are part of IoT networks, the primary difficulty facing IoT technology is the provision of energy-efficient devices and sensors. Power consumption affects the performance of Internet of Things devices. A gadget is deemed more efficient if its power consumption is reduced. IoT devices are an emerging technology with numerous advantages. However, because of the shared device and sensor connectivity, it consumes more energy. Energy efficiency must be incorporated into cross-layer architecture in IoT frameworks [13] [4].

Energy-efficiency Solutions:

By strategically placing nodes, transmission distances between them may be reduced, which would lower the amount of energy needed for communication [44]. Scheduling techniques might reduce interference and maximize frequency channel utilization, thereby increasing energy efficiency. Furthermore, through resource optimization, this integration may reduce the need for new equipment and infrastructure, thereby lowering network costs [44].

Interoperability Issues:

The Internet of Things (IoT) requires interoperability to function well and be scalable because it facilitates effective data exchange and system-to-system communication. Interoperability, however, continues to be a major obstacle in the IoT space, impeding integration and general acceptance across businesses. Because of these interoperability problems, modifying the network across layers is difficult. While cross-layer designs add dependencies that hinder system upgrades and enhancements, a basic tiered approach minimizes complexity by enabling each layer to operate independently [45][46].

Interoperability allows service providers to select from a variety of wireless networks for access. Interoperability across the board is another challenge on the Internet of Things due to the need to handle a vast array of different devices that are divided among different operating systems [47].

IoT and cloud computing combined can help with this problem in part, but interoperability is essential in other situations. Numerous companies also use interoperability to create unique products that are only functional with one another [48]. From the perspective of multiuser networking, a number of power-conscious routing protocols have been proposed to enhance energy efficiency [49].

Interoperability Solutions:

The variety of solutions for a broad range of IoT platforms and devices has significantly increased in recent years. To enable seamless communication and cooperation across diverse IoT devices, interoperability is essential [50]. The article [50] presents novel functionalities in resource trading among roaming IoT devices, trust management, service level agreements, and IoT platform federations in organizational interoperability. To provide design requirements for IoT domain interoperability, many standardization efforts are currently in progress. Furthermore, IoT-EPI efforts are presenting ways to achieve interoperability across various IoT systems by concentrating on interoperability approaches that cover multiple levels of the IoT architecture [50][23]. The figure 6 below shows the cross-layer design interoperability [50]

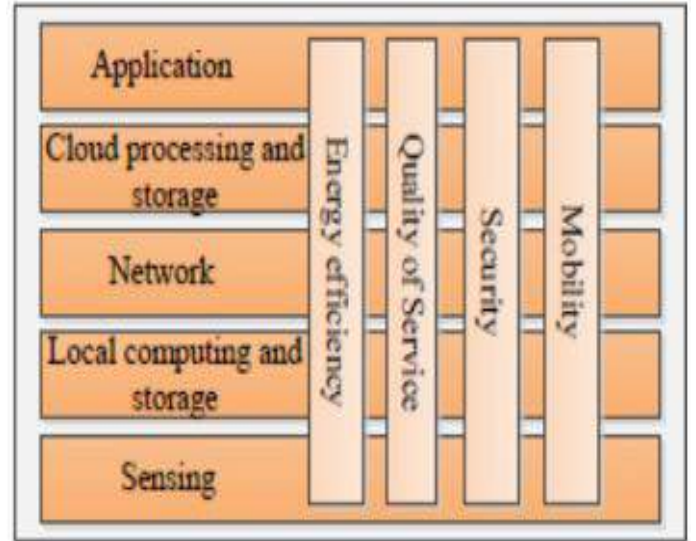


Figure 6. Interoperability in Cross-Layer Design

VI. CONCLUSION

In conclusion, the Internet of Things has grown to become a massive ecosystem with billions of linked objects. Opportunities and problems have arisen because of this expansion, especially in cross-layer design.

A cross-layer design strategy can enhance system performance and ensure robust security in the Internet of Things. IoT devices often collect data via wireless sensor networks; this information is then transmitted to data centers for analysis and decision-making. Energy constraints and standardization issues in IoT devices can hinder the efficacy of the system [51-52].

Although there is much room for improvement in IoT system optimization with cross-layer design, there are a number of trade-offs and challenges involved. To fully benefit from the cross-layer design, obstacles including signaling overhead, scalability limits, and interoperability concerns must be resolved. Furthermore, maintaining safety and compatibility across various IoT platforms and devices continues to be a major

problem that calls for creative solutions and industry standards.

the integration of cross-layer architecture in Internet of Things systems has great promise for improving networked communication's efficiency, dependability, and performance. Mechanisms like mobility optimization and Quality of Service (QoS) management may be effectively applied, enhancing user experiences and system resilience, with coordinated efforts throughout protocol stack levels. Notwithstanding these challenges, investigating cross-layer design in IoT has the potential to spur innovation and build smarter, more interconnected environments across a range of sectors.

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